User Guide

Beyond Mean-Variance Portfolio Optimiser

With Derivatives & Structured Products — A Mental Accounts Framework

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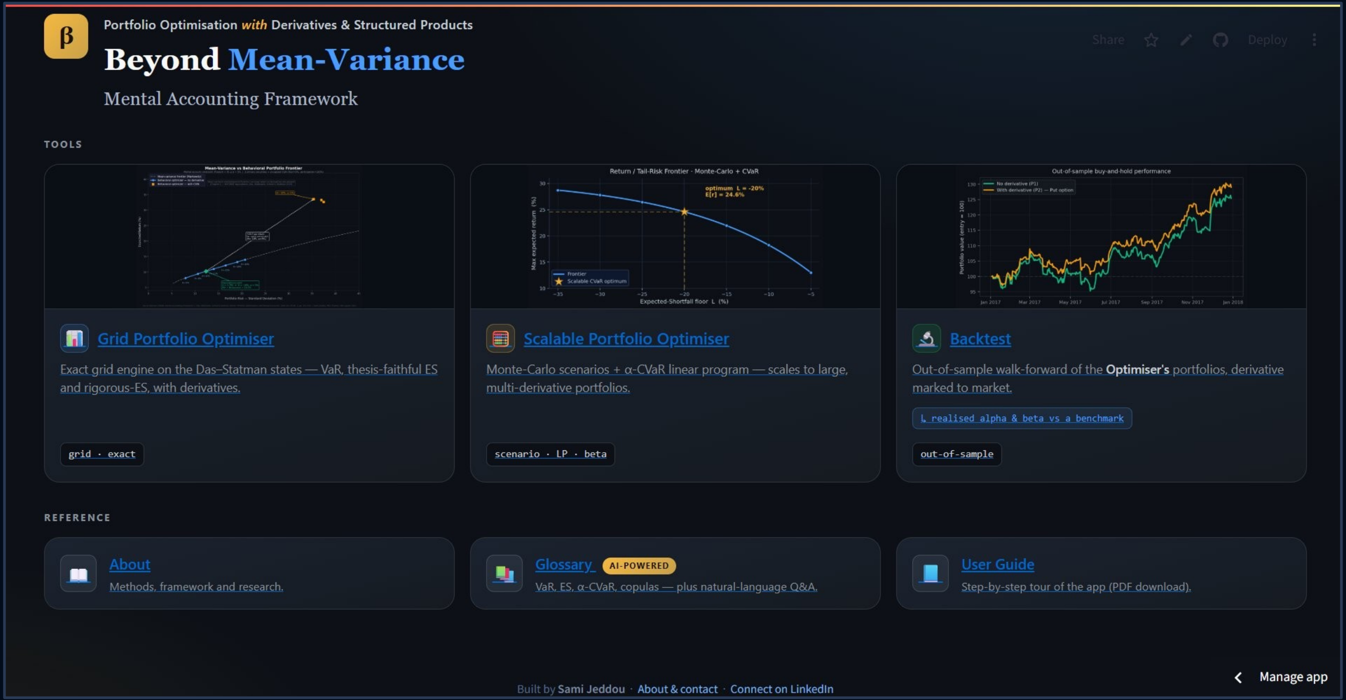
🔗 sami-jeddou-behavioral-portfolio-optimizer.streamlit.app

Home

The landing screen — your starting point: the three tools across the top, with the reference shelf (About · Glossary · User Guide) below.



Home — choose a tool

📊

Grid Portfolio Optimiser

Exact grid engine on the Das–Statman states — VaR, thesis-ES and rigorous-ES, with derivatives.

🧮

Scalable Portfolio Optimiser

Monte-Carlo scenarios + α-CVaR linear program — scales to large, multi-derivative portfolios.

🔬

Backtest

Out-of-sample walk-forward — expected vs realised, plus realised alpha & beta.

Reference & navigation

📖	**About:**	Methods, the mental-accounting framework and the research behind the app.
📚	**Glossary:**	Plain-language, AI-powered definitions — VaR, ES, α-CVaR, copulas, alpha & beta.
💡	**Tip:**	Each tile opens a tool; the Grid Portfolio Optimiser is the place to start.

Grid Portfolio Optimiser — overview

Interactive Frontiers

MV vs Behavioural vs Derivative frontier — visualised side by side on a single chart

Three Portfolios

No derivative / With derivative / Same risk — three perspectives for each optimisation run

AI-Powered

On-demand explanations for every parameter, constraint, derivative type, and result

Portfolio perspectives (0–3)

0	**Portfolio (0) — Markowitz MV optimum:**	Classical mean-variance solution; Portfolio (1) coincides with it.
1	**Portfolio (1) — No derivative:**	Equivalent to MV optimum — confirms MVT/MAT equivalence. Most classical optimisers stop here.
2	**Portfolio (2) — With derivative:**	Same implied λ and downside constraint (H,α or H,α,L) — shows return uplift from derivatives.
3	**Portfolio (3) — Same risk as (1):**	Interpolated at P(1) std dev — direct comparison of return gain at equivalent risk (indicative).

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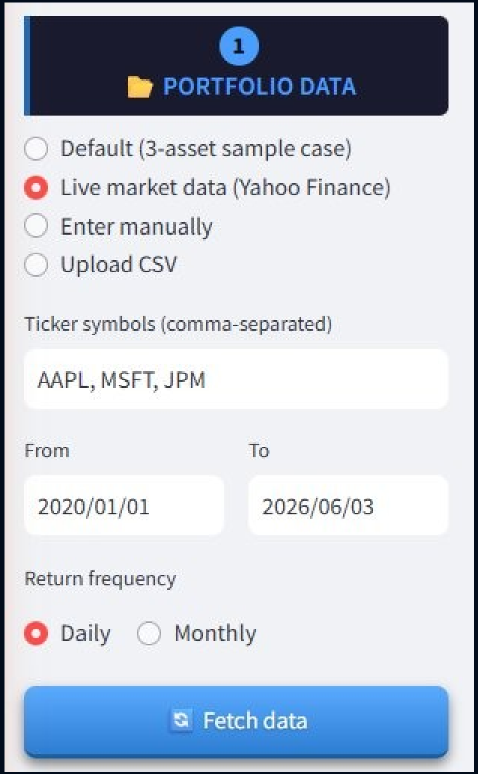
Step 1 — Portfolio Data



Default

Das & Statman (2009) base case — 3 securities, pre-calibrated. Reproduces thesis results.

Live market data

Any global ticker from Yahoo Finance — equities, ETFs, indices, crypto (BTC, ETH...)

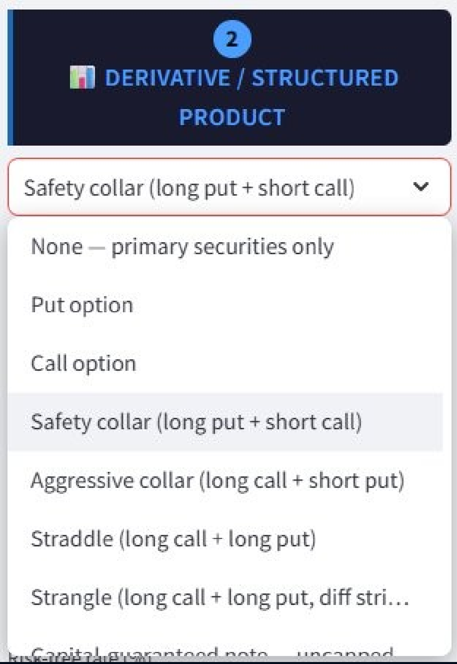
Enter manually

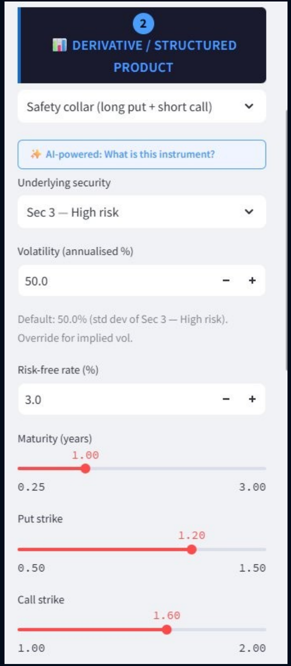
Enter means, std devs, and correlation matrix directly. Supports 2–10 securities.

Upload CSV

First col = dates, remaining cols = asset prices. Returns computed automatically.

Step 2 — Derivative / Structured Product



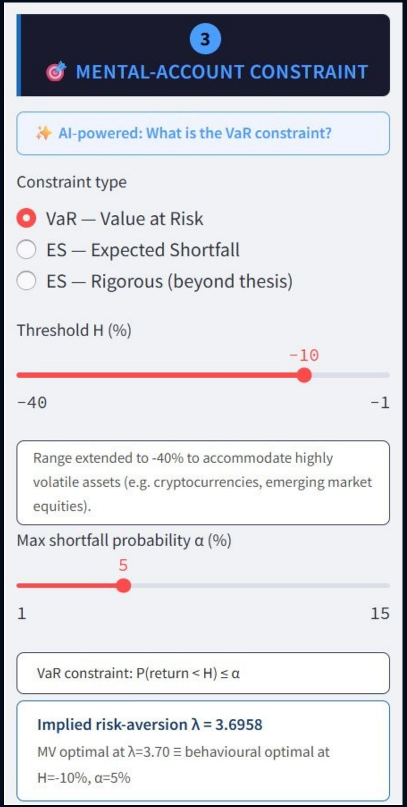


Available instruments

- ▶ Vanilla options — put, call
- ▶ Collars — safety, aggressive
- ▶ Volatility — straddle, strangle
- ▶ Spreads — bull call, bear put
- ▶ Butterfly & condor — long butterfly, call condor
- ▶ Capital-guaranteed note — capped or uncapped
- ▶ Barrier-M note & reverse convertible
- ▶ Discount & outperformance certificates

💡 Select 'None' for no derivatives, or 'Custom' to build your own (calls, puts, digitals, ZCBs).

Step 3 — Mental-Account Constraint



VaR — Value at Risk

$P(\text{return} < H) \leq \alpha$

Caps the probability of finishing below H at α.

ES — Expected Shortfall

$E[\text{return} \mid \text{return} < H] \geq L$

Floors the average tail loss below H at L (thesis).

ES — Rigorous (beyond thesis)

$E[r \mid r < H] \geq L$ *(enforced in the solve)*

Enforces the ES floor in the optimisation itself.

💡 Implied λ is computed dynamically — adjust H and α to see the equivalent risk-aversion coefficient update in real time, confirming MVT/MAT equivalence.

Step 4 — Grid Resolution

⚡ Fast (m=21, m'=15)

⏱️ ~1–2 min (or more depending on the portfolio size)

Quick preview and parameter exploration

⚖️ Standard (m=35, m'=50)

⏱️ ~5–15 min (or more depending on the portfolio size)

Good balance — close to thesis results

🎯 High precision (m=51, m'=99)

⏱️ 1–2 hrs (or more depending on the portfolio size)

Publication-quality replication of thesis results

Also available

🚀 **Turbo — 4th resolution:** High-precision VaR accuracy in ~seconds (VaR-only, ≤4 assets, no derivative).

Rigorous ES: a fixed high-precision grid (m=51); chosen in the Risk-measure step.

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Step 5 — Run the Optimiser

Computation steps

1. Mean-variance frontier — Markowitz sweep (instant)
2. Behavioural frontier (no derivative) — State space, Gaussian copula, grid search, COBYLA
3. Derivative frontier — Same steps extended with derivative payoff (only if derivative selected)
4. Chart rendering + Results computation

Reading the Chart

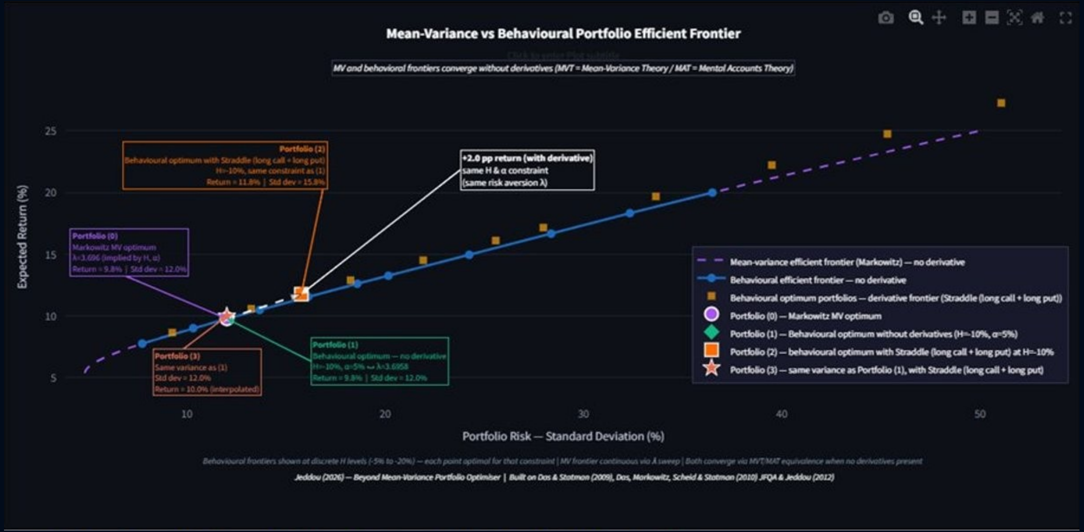


Chart legend

Purple dashed

Mean-variance efficient frontier (Markowitz)

Blue dots

Behavioural efficient frontier — no derivative

Gold squares

Behavioural optimum — derivative frontier

Green diamond

Portfolio (1) — no derivative optimum

Orange square

Portfolio (2) — with derivative, same λ

Coral star

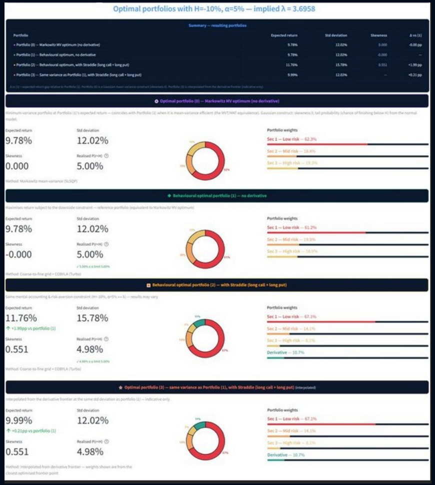
Portfolio (3) — same variance as P(1)

Magenta circle

Portfolio (0) — Markowitz MV optimum

X-axis: Portfolio Risk (Standard Deviation %) | Y-axis: Expected Return (%)

Portfolio Results



Key metrics

Expected return: Annualised mean return of the optimal portfolio

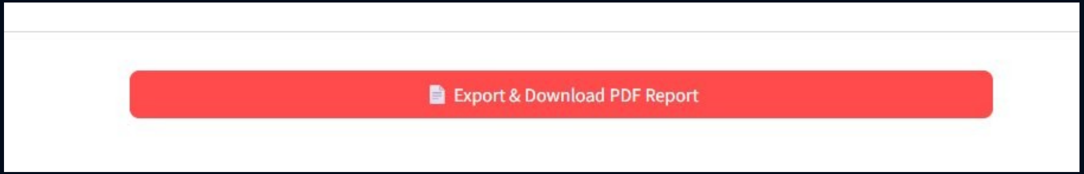
Std deviation: Annualised portfolio volatility (risk measure)

Skewness: Asymmetry of return distribution — positive = upside skew

Shortfall / ES: Expected loss in the tail (ES constraint value)

Portfolio weights: Allocation to each security and derivative

PDF Export



PDF report includes

- ✓ Cover page with run parameters
- ✓ Full efficient frontier chart
- ✓ Portfolio (1), (2), (3) metrics
- ✓ Weight allocation bar charts
- ✓ Legal disclaimer

💡 The PDF is generated in-browser — no data is sent externally. Click 'Download PDF report' after the simulation completes.

Scalable Portfolio Optimiser — Monte-Carlo + CVaR

Scenario engine

Samples thousands of joint return scenarios through a Gaussian or Student-t copula.

α-CVaR linear program

Maximises return subject to a CVaR (Expected-Shortfall) floor — Rockafellar–Uryasev.

Institutional scale

Cost grows linearly in the number of assets — dozens of holdings, several derivatives at once.

How to use it

1	**Pick assets & scenarios:**	Enter tickers and derivatives, choose the copula and the number of Monte-Carlo scenarios.
2	**Set the ES floor:**	Choose the loss level L; the engine sweeps the return / tail-risk frontier.
3	**Read the optimum:**	★ marks the chosen portfolio; weights, E[r], realised ES, vol and skew are reported.

Reading the Scalable output

What you're seeing

- Curve = the return / tail-risk frontier: the most expected return reachable for each Expected-Shortfall floor L.
- A tighter floor (toward −10%) allows less tail loss and lowers return; a looser one (toward −30%) raises it.
- The ★ is the optimum at your chosen floor (here L = −20%, E[r] = 24.6%).
- Built from Monte-Carlo scenarios solved by a CVaR linear program, so it scales to many assets.

Out-of-sample back-test

Two windows

Build weights on a construction window, then buy-and-hold them through a later evaluation window.

Marked to market

The derivative is repriced each step with Black-Scholes, giving a full realised return path.

Alpha & beta

Realised alpha, beta and R^2 of each holding and of the portfolio vs a chosen benchmark.

What you get

1. **Expected vs realised:** Return, volatility and the loss-threshold outcome, for P1 (no derivative) and P2 (with).
2. **Alpha / beta table:** Per-security and portfolio beta (the regression slope) and annualised Jensen alpha, with R^2.
3. **Optional CAPM:** Enter an expected market return E[Rm] to add a CAPM required return and an ex-ante alpha.

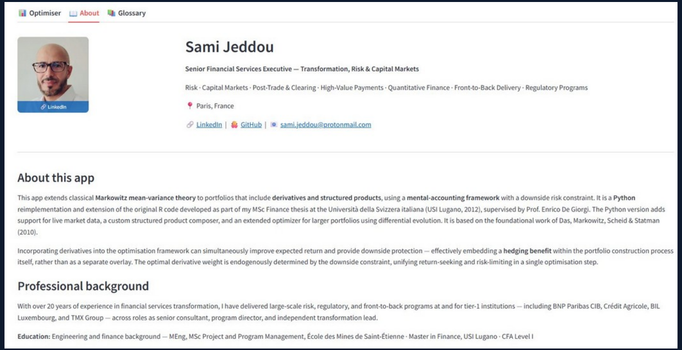
Reading the Backtest output

What you're seeing

- Out-of-sample value of two optimiser portfolios, rebased to 100 at entry and marked to market.
- Green = Portfolio (1), no derivative; orange = Portfolio (2), with the chosen derivative (a put).
- The derivative line cushions drawdowns and ends higher — the realised hedging benefit on unseen data.
- Realised alpha & beta vs a benchmark are computed from these paths (table beneath the chart in the app).

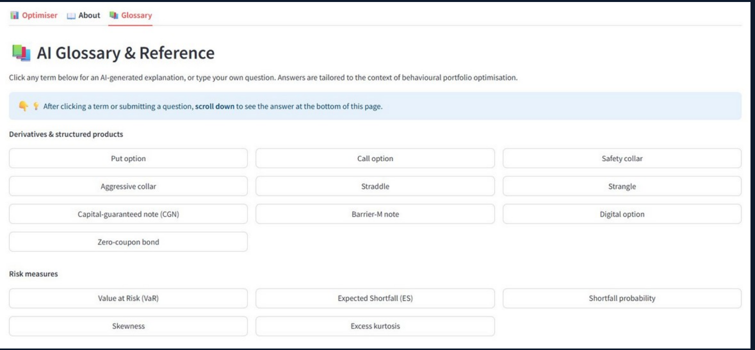
About & Glossary Tabs

📖 About Tab



- ▶ Author background & contact
- ▶ App description & algorithm
- ▶ Academic references
- ▶ Legal disclaimer

📚 Glossary Tab



- ▶ Click any term for AI explanation
- ▶ Covers derivatives, risk, portfolio theory
- ▶ Ask your own questions
- ▶ Tailored to behavioural portfolio context

Disclaimer & Contact

⚠️ Legal Disclaimer

This application is based on the mental accounts portfolio optimisation framework of Das & Statman (2009) and Das, Markowitz, Scheid & Statman (2010), as extended in Jeddou (2012). It is provided for educational and research purposes only and does not constitute financial advice, investment recommendations, or a solicitation to buy or sell any financial instrument. Results are purely illustrative and should not be used as the basis for any investment decision.

Academic References

- Das, S. & Statman, M. (2009) — Beyond Mean-Variance: Portfolios with Derivatives and Non-Normal Returns in Mental Accounts
- Das, S., Markowitz, H., Scheid, J. & Statman, M. (2010) — Portfolio Optimization with Mental Accounts, JFQA Vol.45 No.2
- Jeddou, S. (2012) — Beyond Mean-Variance: Options and Structured Products in Behavioral Portfolios, MSc Finance, USI Lugano

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